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Simulations and Analysis of Adiabatic Quantum Annealing

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Abstract

This project aimed to investigate some properties of a quantum annealer by simulating it on a classical computer. We chose the problem Hamiltonian on the form of the Ising model with interaction energies between all qubits.

The results of our investigation showed at what part of the quantum annealing the minimum energy gap typically occurs for randomly generated Hamiltonians. We saw how the quantum annealing process should proceed in order to achieve a higher probability of obtaining the correct answer. Additionally, we noted that the annealing time increases with qubit count. Interesting phenomenon as “self-rescue” and wave-like convergence arose during the investigation. We also presented an algorithm for solving the game Minesweeper in this paper.

The results in this project can be used to find potential properties of larger systems and also as a base for possible future extensions.



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Sammanfattning

Syftet med detta projekt är att undersöka några egenskaper av en quantum annealer genom att simulera det på en klassisk dator. Vi valde problem-Hamiltonianen i form av Ising modellen med interaktionsenergier mellan alla kvantbitar.

Resultaterna av vår undersökningar visade i vilken del av quantum-annealingen som minsta energigapet typiskt förekommer för slumpmässigt genererade Hamiltonianer. Vi såg hur annealing-processen borde fortskrida i avsikt att uppnå en högre sannolikhet att få korrekt svar. Vi noterade dessutom att beräkningstiden ökade med antal kvantbitar. Intressanta fenomen som "själv-räddning" och våg-liknande konvergens dykte upp under undersökningen. Vår egen algoritm för att lösa spelet Røj är också inkluderad i arbetet.

Resultaterna i detta projekt kan användas till att hitta potentiella egenskaper hos större system och även användas som en bas för möjliga framtida utvidgningar.

Simulations and Analysis of Adiabatic Quantum Annealing

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Chapter 1

Introduction

The term adiabatic quantum annealing (AQA) or quantum annealing (QA) is similar to simulated quantum annealing (SQA), which bears some resemblance to simulated annealing (SA). Both SQA and SA are algorithms that search for the minimum energy in an energy landscape using the idea of annealing, the act of heating something before letting it cool down slowly. The difference between SA and SQA is that the former mimics a natural thermal annealing process while the latter mimics a natural quantum annealing process where tunneling between states is possible. SA and SQA are algorithms to be run on a classical computer, while AQA, or QA, are quantum algorithms to be run on a quantum annealer. In this project we simulate QA on a classical computer and investigate some of its properties. Note that this is different from SQA; we perform a complete simulation of the physical system that is a quantum annealer rather than run an efficient classical algorithm inspired by quantum mechanics. In this project we also explain how the simulation can be done on a classical computer and give some background theory on QA.

A quantum algorithm implemented on a universal quantum computer can be exponentially faster than our best known classical algorithms; Shor's algorithm for prime factorization is an example of this. A quantum annealer however is not a universal quantum computer; it does not have any control of the state other than forcing it to a simple ground state at the beginning of the annealing. Instead the quantum annealer focuses on controlling the Hamiltonian of the state. This can be done by controlling the interaction energies between qubits and controlling magnetic fields which affect the energy of the qubits individually. Adiabatic quantum annealing relies on the adiabatic theorem which states that if the state starts out in the ground state and the Hamiltonian changes slowly enough there is a good probability of the state remaining in the ground state, as long as the ground state is non-degenerate during the annealing process. Some optimization problems can be formulated as a problem Hamiltonian created by the interaction energies and magnetic fields described above so that its ground state represents the solution to the problem.

Whether or not functioning quantum annealers are more efficient than classical computers is debated which we write about in the Background Theory chapter. There are also technical difficulties in producing them; quantum decoherence is one of the main technological difficulties. Despite this, D-wave is producing them and after previously

creating a quantum computer with 1000 qubits they produced one 2017 with twice as many qubits (Gibney 2017).

In our investigation we first looked at how to perform the simulations numerically and how optimization problems can be formulated so that quantum annealers can solve them. We then looked at how the energies of the Hamiltonian changes with time and how this can be used to adjust the speed at which the annealing is performed. We also looked at what happens to the probability of measuring the wanted state when we change the annealing time or change the number of qubits. Throughout the Investigation chapter we present our results and discuss why we get these results and how they can be used when working with a real quantum annealer. Finally, we present our algorithm for solving the game Minesweeper.

Chapter 2

Background Theory

2.1 Adiabatic Quantum Annealing

The Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle \quad (2.1)$$

determines how a state $|\psi(t)\rangle$ evolves over time under the action of a given Hamiltonian $\hat{H}(t)$. For **adiabatic quantum annealing**, the general Hamiltonian consists of two parts: an initial Hamiltonian $\hat{H}_I$ and a final Hamiltonian $\hat{H}_F$.

$$\hat{H}(t) = a(t)\hat{H}_I + b(t)\hat{H}_F \quad (2.2)$$

$$a(0) = b(t_F) = 1 \quad (2.3)$$

$$a(t_F) = b(0) = 0, \quad (2.4)$$

so that $\hat{H}$ starts off as the initial Hamiltonian $\hat{H}_I$ and ends up as the final Hamiltonian $\hat{H}_F$. The functions $a(t)$ and $b(t)$ can be chosen arbitrary. As an example, they can be chosen as linear functions of time

$$a(t) = 1 - \frac{t}{t_F} \quad (2.5)$$

$$b(t) = \frac{t}{t_F}. \quad (2.6)$$

The adiabatic quantum annealing process is as follows: the **ground state** of $\hat{H}_I$ is chosen to be the initial state. The system then evolves in time with the time-dependent Hamiltonian $\hat{H}(t)$ and at the end of the annealing process, the system is in some final state. If the ground state energy of $\hat{H}(t)$ is **non-degenerate** throughout the annealing process, the system certainly will be in the **ground state of $\hat{H}(t = t_F) = \hat{H}_F$ at $t = t_F$** , provided that the annealing process is **infinitely slow** (Griffiths 2013). In reality, because of the fact that no annealing process can take infinitely long time, there will be an **uncertainty in the outcome**. The condition of non-degenerate energies can have an exception which is **degenerate ground state energy of $\hat{H}_F$** . This is not a problem if we do not care about which of the degenerate ground states the system ends up with as long as it is in a minimal energy configuration.

There is a trick to choosing $\hat{H}_I$ and $\hat{H}_F$: to have them act in orthogonal directions. There are two reasons behind this choice. First, with the ground state of $\hat{H}_I$ as the initial state, the orthogonality ensures that this state can be expressed as an equiprobable superposition of all basis vectors in the direction of $\hat{H}_F$ (Mcgeoch 2014). Second, since the function $a(t)$ is not zero until the end of the annealing process, the initial Hamiltonian $\hat{H}_I$ will always be a part of the total Hamiltonian $\hat{H}$ except at the very end, which makes energy degeneracy less likely to occur. Also, since the system starts off as the ground state of $\hat{H}_I$, the initial Hamiltonian should be chosen to give an initial state that is easy to prepare. The final Hamiltonian $\hat{H}_F$ should correspond to the problem that we want to solve and the final ground state is the solution to the problem.

2.2 The Adiabatic Approximation

A general way to represent the state $\Psi(t)$ is

$$\Psi(t) = \sum_n c_n(t) \psi_n(t) e^{i\theta_n(t)}, \quad (2.7)$$

where

$$e^{\theta_n(t)} \equiv e^{-\frac{i}{\hbar} \int_0^t E_n(t') dt'} \quad (2.8)$$

is a phase factor at time t ; $E_n(t)$ and $\psi_n(t)$ is the n th time-dependent eigenvalue and eigenfunction of the time-dependent Hamiltonian at time t . As conventional, $|c_n(t)|^2$ is the probability of measuring the eigenstate $\psi_n(t)$ at time t . Eq. (2.7) inserted in the Schrödinger equation leads to

$$\dot{c}_m(t) = -c_m \langle \psi_m | \dot{\psi}_m \rangle - \sum_{n \neq m} c_n \frac{\langle \psi_m | \dot{H} | \psi_n \rangle}{E_n - E_m} e^{-\frac{i}{\hbar} \int_0^t [E_n(t') - E_m(t')] dt'}, \quad (2.9)$$

which is an exact result (Griffiths 2013). For a system with a time-dependent Hamiltonian that changes infinitely slow, the second term in Eq. (2.9) can be set to zero, so that if $|\Psi(t=0)\rangle = |\psi_0(t)\rangle$ then $|\Psi(t_F)\rangle = |\psi_0(t_F)\rangle$. As mentioned previously, neither numerical solutions or any real physical process takes infinitely long time, which means that this term will have some effect.

We want to observe how the ground state probability changes with time, that is, $\frac{d}{dt} |c_0(t)|^2$. The denominator, $E_n - E_0$ in Eq. (2.9), implies that the difference between the n th and the ground state energy comes into play: the smaller energy difference, the bigger $\dot{c}_0(t)$. Since the sum is over all n except zero, all energy values of the excited states will affect the time derivative of the ground state probability. Moreover, the exponential factor shows that $\dot{c}_0(t)$ not only depends on the size of the energy gaps at the present time t , but also on all historical values from beginning of the process up to time t . Another thing to be noticed is the time derivative of Hamiltonian in the nominator; the faster the Hamiltonian varies, the bigger $\dot{c}_0(t)$ becomes. However, the relationship between $\frac{d}{dt} |c_0(t)|^2$ and $\dot{c}_0(t)$ is not simple since it depends on the phase of the ground state.

2.3 Single Qubit State Representation

The states of a single qubit lives in a two dimensional complex vector space, with $|0\rangle$ and $|1\rangle$ as its basis. Consider the case with electron spin directions as a physical two-level system. The basis states of a qubit are conventionally represented in vector form as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad (2.10)$$

for the spin up state and

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (2.11)$$

for the spin down state.

2.4 Multiple-Qubit State Representation

To represent multiple-qubit states, the tensor product is used to form vector spaces of higher dimension. The abstract definition of tensor product has a matrix representation, known as the Kronecker product, as follows: suppose A is an m by n matrix and B is a p by q matrix, the tensor product in matrix representation is

$$A \otimes B \equiv \begin{bmatrix} A_{11}B & A_{12}B & \dots & A_{1n}B \\ A_{21}B & A_{22}B & \dots & A_{2n}B \\ \vdots & \vdots & \vdots & \vdots \\ A_{m1}B & A_{m2}B & \dots & A_{mn}B \end{bmatrix}, \quad (2.12)$$

where all terms $A_{ij}B$ denote p by q submatrices whose entries are proportional to B , with the overall proportionality constant A_{ij} (Nielsen and Chuang 2000).

By applying the tensor product according to Eq. (2.12), the two-qubit state $|\Psi\rangle = |\psi_1\psi_2\rangle = |01\rangle$, representing a system with the first electron spin up and the second electron spin down, becomes as follows

$$|01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}. \quad (2.13)$$

The basis vectors of any N-qubit system can be obtained using the same method.

Using equations (2.10), (2.11) and (2.12) the arbitrary state for two qubits can be written as follows. The vector in the middle of the equation uses our state convention which is explained using regular vectors in the scalar multiple in the right hand side of the equation.

$$a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} * \begin{pmatrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{pmatrix}. \quad (2.14)$$

Notice in Eq. (2.14) that the ket state index in the vector to the right hand side of the scalar multiple are ordered by binary size. This will always be the case regardless of the number of qubits. This can be used to create a general representation of the state as

$$|q, N\rangle = |s_N\rangle \otimes |s_{N-1}\rangle \otimes \cdots \otimes |s_1\rangle, \quad (2.15)$$

where q is the decimal representation of the binary integers $s_N s_{N-1} \dots$ and N is the number of qubits used.

2.5 Tensor Product of Linear Operators

By definition of the tensor product, if C and D are linear operators in vector spaces V and W , respectively; $C \otimes D$ is a linear operator that can act on all elements in the vector space $V \otimes W$. In mathematical language, it is (Nielsen and Chuang 2000)

$$C|v\rangle \otimes D|w\rangle \equiv (C \otimes D)(|v\rangle \otimes |w\rangle). \quad (2.16)$$

In a 2^N -dimensional complex vector space, the Pauli matrices will all have the dimension $2^N \times 2^N$. As an example, the 2-dimensional Pauli-x matrix and Pauli-z matrix in the z-basis is

$$\sigma^x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma^z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (2.17)$$

The tensor product of Pauli matrices σ^x and σ^z , each acting on their respective two dimensional complex vector space is, according to definitions (2.12) and (2.16)

$$\sigma^x \otimes \sigma^z = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}. \quad (2.18)$$

2.6 Linearly Time-varying Hamiltonian of the Ising Model

An electron with angular momentum forms a circulating current and thus a magnetic moment. The magnetic moment $\vec{\mu}_L$ due to orbiting angular momentum $\vec{L}$, is given by the formula $\vec{\mu}_L = -\frac{e}{2m_e}\vec{L}$ (Harris 2013), where e and m_e denotes the charge and the mass of an electron. Electrons also have an intrinsic angular momentum, called spin, $\vec{S}$, which gives an intrinsic magnetic moment, $\vec{\mu}_S$, given by the formula $\vec{\mu}_S = -\frac{e}{m_e}\vec{S}$ (Harris 2013). In a real physical system, magnetic moment of both types will interact with an externally applied magnetic field. Besides these, there will also be magnetic moment contributions from the atomic nucleus. For the sake of computational simplicity and also to avoid straying from the concept of quantum annealing, only the intrinsic magnetic moment is considered. Here and after, an upward pointing $\vec{\mu}_S$ is called as “spin” up and a downward pointing $\vec{\mu}_S$ is called as “spin” down. The reason for this seemingly arbitrary notation is that a physical implementation of a quantum annealer does not necessarily use magnetic

moment or spin, any two-level system works.

The Ising model for ferromagnetism is named after Ernst Ising, whom, together with his professor Wilhelm Lenz, developed the model in 1924. The model describes a group of N particles, in our case, N qubits, each being affected by an external magnetic field and the spin of every other qubit (Selinger 2016). The total energy of the system, H_F , due to all interaction energies is thus

$$H_F = - \sum_i h_i^z s_i^z - \sum_i \sum_{j>i} J_{ij} s_i^z s_j^z, \quad (2.19)$$

where s_i^z and s_j^z denotes the spin in the z-direction of qubit number i and j ; h_i^z is the strength of the externally applied parallel magnetic field in z-direction at electron number i ; J_{ij} is the interaction energy between spin i and j . The first part of the total energy has a negative sign because magnetic moment parallel to external magnetic field have a minimum potential energy U , due to the formula $U = -\vec{\mu} \cdot \vec{B}$. The second part of the total energy have a negative sign because of the nature that neighboring magnetic moments tends to be in a parallel alignment in the case of ferromagnetism. As a consequence, the coefficients J_{ij} should be chosen as positive. On the other hand, for anti-ferromagnetic materials, J_{ij} has to be negative. More generally, we may consider the case where some J_{ij} are positive and some negative. The system will then be magnetically frustrated which means that it will be very hard to find the ground state configuration(s) by using a conventional computer. Eq. (2.19) can be easily generalized to the following operator form

$$\hat{H}_F = - \sum_i h_i^z \sigma_i^z - \sum_i \sum_{j>i} J_{ij} \sigma_i^z \sigma_j^z. \quad (2.20)$$

As mentioned in chapter 2.1, the initial Hamiltonian should be chosen to give an initial state that is easy to prepare. This can be done by applying a large transverse magnetic field in the x-direction and waiting for the system to arrive at its ground state. The initial Hamiltonian in operator form is thus

$$\hat{H}_I = -h^x \sum_i \sigma_i^x, \quad (2.21)$$

where h^x is the strength of the externally applied transverse field and σ_i^x is the Pauli-x matrix acting on the spin in the x-direction of qubit number i . Taken together, the linearly time-varying Hamiltonian of the Ising model is

$$\hat{H}(t) = (1 - \frac{t}{t_F}) \hat{H}_I + (\frac{t}{t_F}) \hat{H}_F = (1 - \frac{t}{t_F}) (-h^x \sum_i \sigma_i^x) + (\frac{t}{t_F}) (\sum_i -h_i^z \sigma_i^z - \sum_i \sum_{j>i} J_{ij} \sigma_i^z \sigma_j^z). \quad (2.22)$$

2.7 Quantum Speedup

Researchers are currently investigating “quantum speedup” in quantum annealers (Rønnow et al. 2014). One group defines quantum speedup as the ratio

$$S(N) = \lim_{N \rightarrow \infty} \frac{Q(N)}{C(N)}, \quad (2.23)$$

where N is the size of the problem, $C(N)$ is the time for a classical device or algorithm and $Q(N)$ is the time for a quantum device. $S(N)$ is the quantum speedup. $C(N)$ can be either chosen to be the time using the best *known* classical algorithm, or the best *possible* classical algorithm. An example of quantum speedup is Grover's search algorithm which is of complexity $\mathcal{O}(\sqrt{N})$ on a universal quantum computer while it has been proved that the best possible classical algorithm has the complexity $\mathcal{O}(N)$. However, the best possible classical algorithm is unknown for many interesting problems. For Grover's algorithm or other very efficient quantum algorithms it might be more suitable to use a definition of quantum speedup with a logarithm

$$S(N) = \lim_{N \rightarrow \infty} \frac{\log Q(N)}{\log C(N)}, \quad (2.24)$$

since otherwise we would simply get 0.

Due to the nature of quantum mechanics there is always a non-zero probability that the quantum annealer will give an incorrect answer. This can make it hard to define the complexity of the annealer, since no matter how long it takes, it will never give the correct answer with the probability 100 %. The solution to this problem is to choose a success probability which is sufficient, for example 99 %, and to measure the time it takes to reach the correct answer to that probability. Instead of only investigating the probability of finding the correct answer (the state with the lowest cost) it is possible to also look at the expectation value of the cost. Which one of these measures that are the most interesting depends on the situation.

Grover's search algorithm cannot be run efficiently on a quantum annealer and it can be difficult to find a problem where quantum annealing gives a quantum speedup. There has been findings of a quantum speedup of QA for certain constructed problems compared to simulated annealing (SA) or simulated quantum annealing (SQA). However, other better classical algorithms might exist, and for some problems they were even expected to exist.

QA is often compared to SA (Denchev et al. 2016). SA can struggle with finding the lowest energy when the energy landscape has many deep minimums that are close together. For this type of energy landscape QA can be exponentially faster than SA. However, SQA can scale equally well as QA on a D-wave 2X device for one such problem. They also expected other classical algorithms to be even more efficient. This was because the D-wave 2X device can not simulate the Ising energy model due to lacking connections between qubits. Engineering advanced connectivity graphs is an active field of research and is a very difficult task. Even though the D-wave 2X did not give a scaling advantage over SQA, it was still 10^8 times faster for the problem they investigated. Since the time factor of 10^8 for D-wave 2X compared to SQA was not compared to the best classical algorithm this team did not prove quantum speedup.

Additionally, they wanted to emphasize that a speedup for non-infinite problem sizes was very much of interest as well. This increase in speed can be hidden using the definition of quantum speedup given in Eq. (2.23). The reason for this was the big factor 10^8 , but also the fact that this factor has increased with improved technology from 10^4

on the D-wave Vesuvius chip, and can continue to increase.

It is important to continue investigating quantum speedup in QA to learn about which types of problems a quantum annealer might be useful for and how they can be solved. Furthermore, if quantum annealers are found to be much more efficient than all classical algorithms for an important problem then that incentive can speed up the development of better quantum annealers.

Chapter 3

Investigation

To simulate a quantum annealer on a classical computer, we used the multiple-qubit state representation presented in section 2.4 and figured out the general matrix form of Pauli-x matrices and Pauli-z matrices acting on qubit number i . Two different numerical methods are then implemented in MATLAB to solve the Schrödinger equation. This allowed us to simulate the quantum annealer. Our next step was to look at how optimization problems can be rewritten to the Ising-form. With these basics down we began the investigation of the properties of a quantum annealer. This was done by first examining the energy spectrum of randomly generated Hamiltonians. We then looked at how the ground state of the annealing changes during the process and how the rate of change of the Hamiltonian can be changed to improve the performance. Our next step was to investigate how the success of the annealing depended on the running time. We also investigated how annealing time scales with the number of qubits and if this was different for a specific optimization problem. Finally we wrote an algorithm for solving the game Minesweeper.

3.1 Numerical Implementation in MATLAB

3.1.1 Creation of Pauli Matrices

In order to create the Hamiltonian for the Ising model we created the matrices σ_j^z and σ_j^x used in equations (2.20) and (2.21). As described above the index j refers to which qubit the Pauli matrix operates on. We see from the vector representation of the states (2.10) and (2.11) and the Pauli matrices (2.17) that

$$\begin{aligned}\sigma^z|0\rangle &= |0\rangle \\ \sigma^z|1\rangle &= -|1\rangle \\ \sigma^x|0\rangle &= |1\rangle \\ \sigma^x|1\rangle &= |0\rangle.\end{aligned}\tag{3.1}$$

We propose here an algorithm that finds σ_j^z and σ_j^x and show as an example how σ_2^z looks for two qubits. Using the fact that the states are ordered by binary size in Eq.

(2.14) we can write out an arbitrary state in this form. We then operate with the Pauli matrix on this state using equations (3.1). The two qubit example is shown below

$$\sigma_2^z \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} * \begin{pmatrix} \sigma_2^z |00\rangle \\ \sigma_2^z |01\rangle \\ \sigma_2^z |10\rangle \\ \sigma_2^z |11\rangle \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} * \begin{pmatrix} |00\rangle \\ -|01\rangle \\ |10\rangle \\ -|11\rangle \end{pmatrix} = \begin{bmatrix} a \\ -b \\ c \\ -d \end{bmatrix}. \quad (3.2)$$

The final step is to look at the left hand side and the right hand side of Eq. (3.2) to identify σ_2^z

$$\sigma_2^z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}. \quad (3.3)$$

σ_j^z is always diagonal with elements 1 or -1. Finding σ_j^x can be done with the same method. In that case the states will change which means that the matrix σ_j^x will not be diagonal.

3.1.2 Solving the Schrödinger Equation

A simple and straightforward method for numerically solving the Schrödinger equation is the forward Euler method. The general Hamiltonian in Eq. (2.22) consists of one operator that acts in x-direction and one operator that acts in z-direction. Since σ^x and σ^z does not commute, one of the Pauli matrices has to be non-diagonal. For a N-qubit system, the general Hamiltonian will also be a non-diagonal matrix with dimension $2^N \times 2^N$, which will put an early restriction on how many qubits can be used, somewhere around 10 on a regular computer. Another deficiency with the forward Euler method is that the state does not stay normalized so we have to normalize the state vector after each discrete time step.

A more elegant method that both preserves the normalization and avoids regular matrix multiplication is by using $DHDHD$ form, where D stands for diagonal matrix and H stands for Hadamard matrix. This form emerges from a modified version of the Trotter formula, see Appendix A for derivation details. In this method, all Pauli matrices are expressed in their eigenbasis, which means that all of them are now diagonal matrices. This is the reason behind the letter D . To deal with the problem of non-commuting basis, a Hadamard matrix, standing for the letter H , which serves as a basis change matrix, is inserted between Pauli matrices of orthogonal directions. The numerical steps for this method are as follows: as usual, the system starts off as the ground state of $\hat{H}_I$, expressed in the z-basis. A diagonal matrix in the z-basis acts on this initial state under time interval $\Delta t/2$, followed by a Hadamard matrix that changes the state to x-basis. It is then acted on by a diagonal matrix in the x-basis under twice the time duration after which another Hadamard matrix changes the state back to z-basis for finally being again acted on by a diagonal matrix in the z-basis under time interval $\Delta t/2$. After these five steps, the Hamiltonian changes and this procedure starts over again. It continues until the time t_F is reached and the system is then in the final state.

By using an effective algorithm, multiplication of Hadamard matrix with a column vector can be done with complexity $\mathcal{O}(2^N \log 2^N)$, while multiplication of an arbitrary non-diagonal matrix with a column vector, which is the type of operations occurring in the forward Euler method, takes $\mathcal{O}((2^N)^2)$ (Lu and Desmedt 2016). As usual, N denotes the number of qubits. The $DHDDH$ form is more useful for simulations when many qubits are used. In situations where the energy spectrum of $\hat{H}(t)$ needs to be plotted, the forward Euler method becomes more useful, due to the improved numerical complexity, $\hat{H}(t)$ is never truly created as matrices in the $DHDDH$ form in MATLAB, which makes it difficult to extract the eigenvalues of it. Therefore, for the investigations that follow, both numerical methods are used in different situations.

3.1.3 Dealing with Numerical Parameters and Constants

To see that our simulated quantum annealer was functioning properly we ran the simulations with decreasing Δt until decreasing it further only changed the final state insignificantly. We then verified that we could get a higher probability of measuring the ground state when increasing t_F as expected. We also noted that the annealing time is proportional to $\hbar$ and inversely proportional to the Hamiltonian as seen in the Schrödinger equation (2.1). Later in the investigations we randomized h_i^z and J_{ij} with a randomized size proportional to h^x which made the entire Hamiltonian proportional to h^x .

3.2 Rewriting Optimization Problems to Ising Form

It is not always obvious how a problem can be solved using minimization of energy on the Ising form seen in Eq. (2.20). Technological challenges also put restrictions on the Ising model in different ways; the energy constants h_i^z and J_{ij} will have limited maximum values and a certain inaccuracy. Additionally it is hard to make interaction energies J_{ij} between all qubits. A model that takes these aspects into account make it even harder to create constants h_i^z and J_{ij} so that the minimum energy corresponds to a solution of the problem at hand.

We used an article where difficult optimization problems were rewritten to the Ising form (Lucas 2014). A simple example of a problem Hamiltonian for the number partitioning problem is given here. The problem is to separate a single set of numbers into two sets with an equal sum. If the set of numbers is given by $n_1, n_2, \dots, n_i, \dots, n_N$ the Hamiltonian that they suggest which requires N qubits is

$$H_F = A \left(\sum_i^N n_i s_i \right)^2, \quad (3.4)$$

where s_i is 1 for spin up and -1 for spin down just like we used it before. Note that the problem Hamiltonian H_F in Eq. (3.4) is always positive and is 0 if and only if the sum is 0. The sum is 0 if and only if the sums of the sets with the separated numbers are equal. In other words the lowest value of H_F gives the solution to the problem.

With the Hamiltonian on the form given in Eq. (3.4) we now know that it solves the problem, but it is not written exactly on Ising form yet. Expanding the square gives

$$H_F = A\left(\sum_i^N n_i s_i\right)^2 = A\left(\sum_i n_i^2 s_i^2 + \sum_{i \neq j} n_i n_j s_i s_j\right). \quad (3.5)$$

The first term in the right hand side of Eq. (3.5) is simply a constant since s_i^2 is always 1. Minimizing H_F gives the same solution as minimizing $H_F - C$ where C is a constant. This means we can remove this term from the equation and still reach the correct answer. The second term decides the interaction energies $J_{ij} = 2An_i n_j$ for $j > i$ and 0 otherwise. With this choice of J_{ij} we reached the Ising Form in Eq. (2.19). Notice how h_i^z are all chosen to be 0; this can be a practical advantage. The Hamiltonian can be rewritten by forcing the first number to be put into the “first” set, replacing s_1 by 1 in Eq. (3.4). This will decrease the number of qubits required by one.

3.3 Eigenspectrum

The eigenspectrum of the annealing process is a plot of the eigenvalues of the Hamiltonian as a function of time. The eigenspectrum is of importance because of the $E_n - E_m$ term used for non-perfect adiabatic quantum annealing seen in Eq. (2.9). The ratio between H_F and H_I affects the eigenspectrum; the energies at the start will be proportional to H_i and the energies at the end of the annealing process will be proportional to H_F . Because of this we keep the expected absolute values of the Ising model constants h_i^z and J_{ij} equal to the h^x value in equations (2.20) and (2.21). This will create a similar interval of eigenvalues at the start and at the end of the annealing process, at least for a lower number of qubits.

A typical eigenspectrum is show below in Figure 3.1. Here the values of h_i^z and J_{ij} were randomized. The eigenspectrum seen in Figure 3.1 depends on the exact randomized values, but we often see fairly smooth curves that tends to avoid degeneracy.

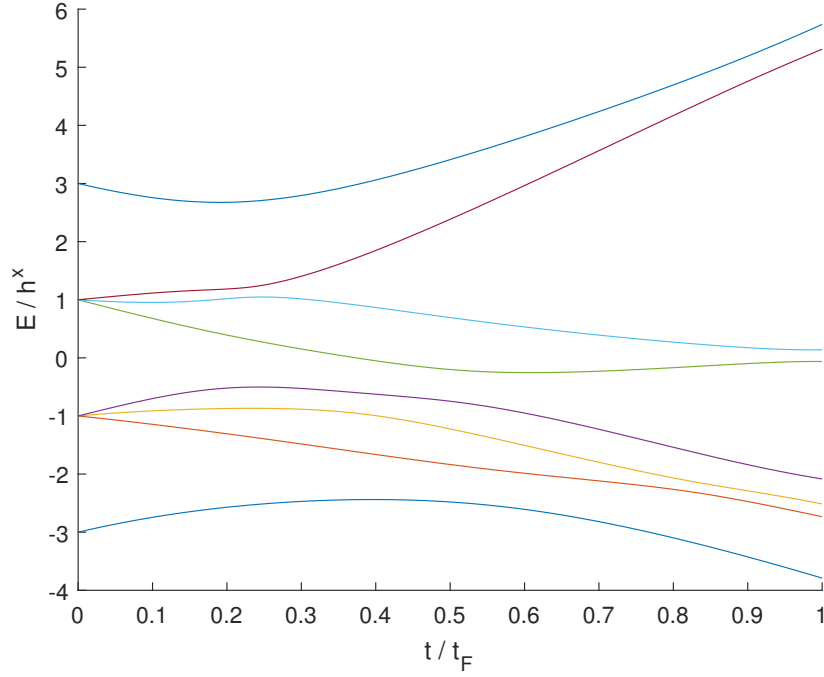


Figure 3.1: A typical eigenspectrum. E is the eigenvalue for each state and h^x is taken from Eq. (2.21).

The lowest energy gap (between the two lowest energies) is of special interest. We note that this gap is independent of H_F at the start of the anneal since $H(t=0) = H_I$. The energy gap at the end depends only on H_F and will sometimes be much smaller than the gap at the start of the annealing process. It is however very rare that the gap will be significantly larger at the end compared to the beginning of the anneal with our chosen ratio between the constants in H_I and H_F .

We wanted to find at which time of the annealing the energy gap between the two lowest eigenvalues typically reaches a minimum. For this we used the constant rate of change in Hamiltonian shown in equations (2.2), (2.5) and (2.6). We set the number of qubits to 4 and randomized the values of h_i^z and J_{ij} 10^5 times with uniform distribution with average value 0, and again 10^5 times with normal distribution with average value 0. As above we kept the expected absolute value of the constants equal to h^x . We ran this twice to get a feel for the accuracy. The histogram below shows at what part of the annealing the minimum gap tends to be located. The time at which the gap is minimal we call t_{g-min} .

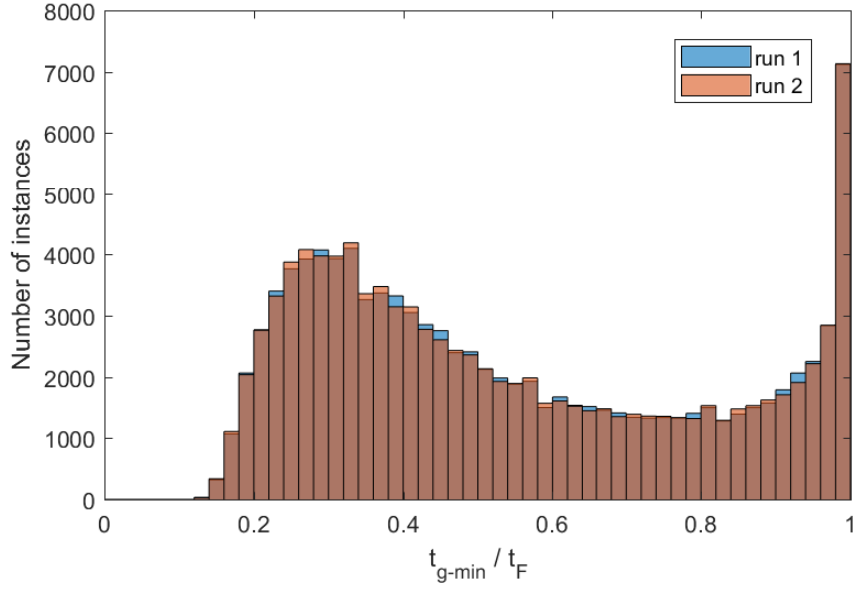


Figure 3.2: Uniform distribution of h_i^z and J_{ij} elements

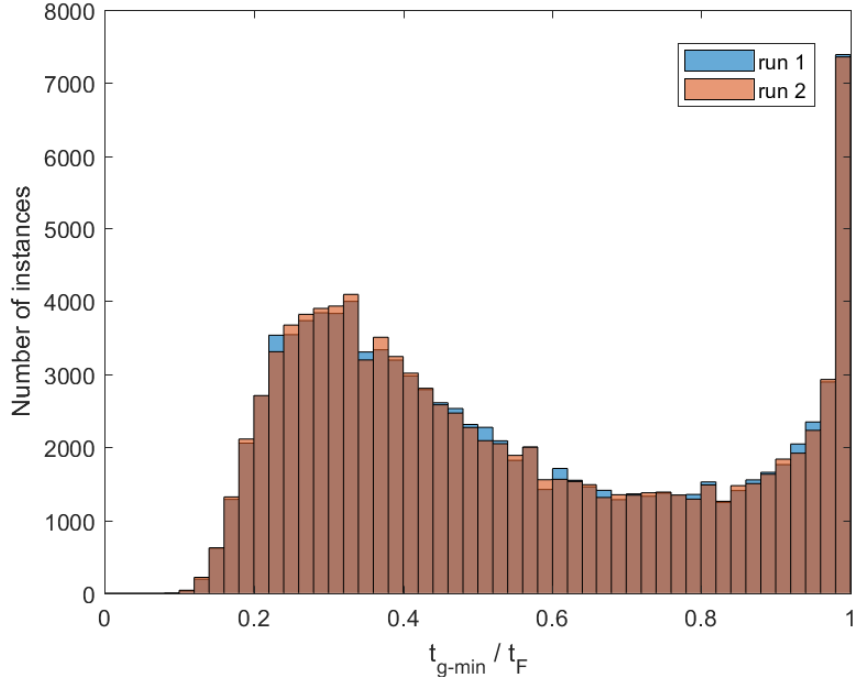


Figure 3.3: Normal distribution of h_i^z and J_{ij} elements

We note from Figures 3.2 and 3.3 that the position of the minimum was very similar for uniform and normal distribution of energy constants h_i^z and J_{ij} . Other than at the first 15% of the total time the gap can be located anywhere with fairly similar probability, with a slight dip in probability between 50% and 90% of the total time. Finally note the big spike at the end of the annealing process of which we believe corresponds

to a monotonically decreasing gap, maybe when the lowest energy gap of H_F is very small.

Because the gap minimum depends more on the exact constants h_i^z and J_{ij} rather than their distribution we used only normal distribution when randomizing them in upcoming investigations.

3.4 Nonlinear Annealing Path

In section 2.2, we noted that for annealing processes with finite execution times, the time derivative $\dot{c}_0(t)$ is bigger when the size of energy gaps between the ground state energy and the excited state energies are smaller; and the time derivative is smaller when the Hamiltonian varies slower. This observation gives a clue of how the annealing process should progress: when the energy gaps are smaller, the Hamiltonian should move forward, from $\hat{H}_i$ towards $\hat{H}_F$, at a slower pace, to make sure that the system stays in the ground state of $\hat{H}$. On the other hand, when the energy gaps are bigger, we might change the Hamiltonian faster with a low risk of jumping up to any excited state. In other words, instead of using a linear annealing path from $t = 0$ to $t = t_F$, it might be better to use a nonlinear path where the progression speed at different times is adjusted to the size of energy gaps. The analysis turns out to be difficult when considering all excited state energies, therefore only the energy gap between the ground state energy and the first excited state energy will be considered when adjusting the progression speed since this produces the dominant term.

Figure 3.4 shows an energy gap that, roughly speaking, decreases with time. The x-axis in this figure represents a scaled total annealing time, $\frac{h^x t_F}{\hbar}$, instead of just the annealing time t_F , to avoid the incredibly small numbers that arises from the reduced Planck constant $\hbar$ otherwise; h^x is a constant discussed in section 3.1.3 above. Figure 3.5 shows how the ground state probability evolves over time for this linearly-varying Hamiltonian. It can be read from this figure that the ground state probability starts off as 1.0, as it should, oscillates a bit until it arrives at a scaled time around 8 unit, where the probability strictly drops down and ends at a value somewhere around 0.65.

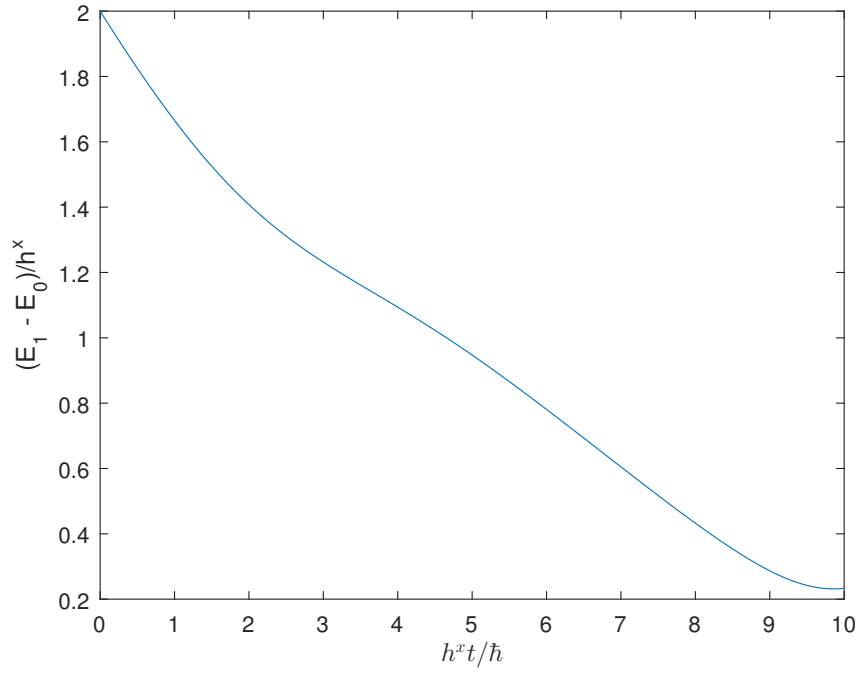


Figure 3.4: Energy gap between ground state and first excited state plotted against scaled time, from start to end.

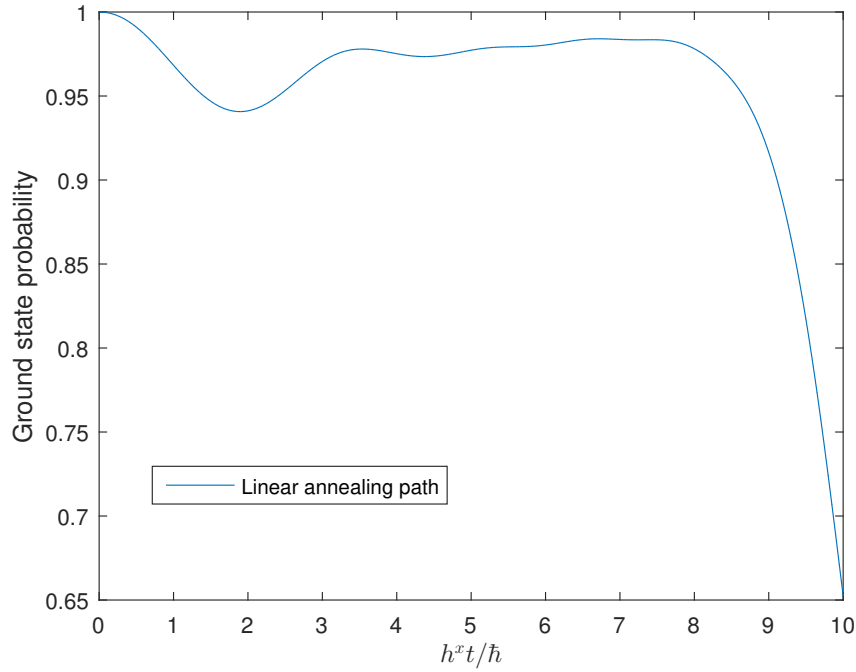


Figure 3.5: How ground state probability evolves with scaled time.

Next, the annealing speed is adjusted by dividing the same total execution time, which is 10 units long, into 5 equal time intervals. The speed relation between the intervals is

chosen as 5, 4, 3, 2, 1. This means that the annealing process runs 5 times as fast in the first interval as in the last interval; it runs 4 times as fast in the second interval as in the last interval but only $\frac{4}{5}$ as fast as in the first interval, etc. The same Hamiltonian and total time was used. The result is shown in Figure 3.6. This time the probability starts off as 1.0 and drastically drops down in the first interval, due to the rapid pace at which the annealing process goes in this interval. At the scaled times between 3 and 7.5 units, the probability, though it is oscillating, keeps at a quite constant level. This might imply that the speeds in this interval were chosen appropriately. In the last interval, where the energy gap becomes very small, the probability again begins to decrease and ends somewhere around 0.755. Comparing Figures 3.5 and 3.6, it can be seen that the probability drop in the last time interval is much less drastic for the non-linear annealing path than the linear annealing path, due to the reduced pace. Even though the action of scaling annealing speed to an overall non-linear form leads to a higher drop of the ground state probability at the beginning of the process, the system is more likely to be in the ground state at the end of the annealing process for the non-linear path.

Figure 3.7 is for the speed relation: 20, 15, 10, 5, 0.5. The last interval in this plot illustrates the fact that when the annealing process runs at a slow enough speed, the ground state probability can almost be held constant.

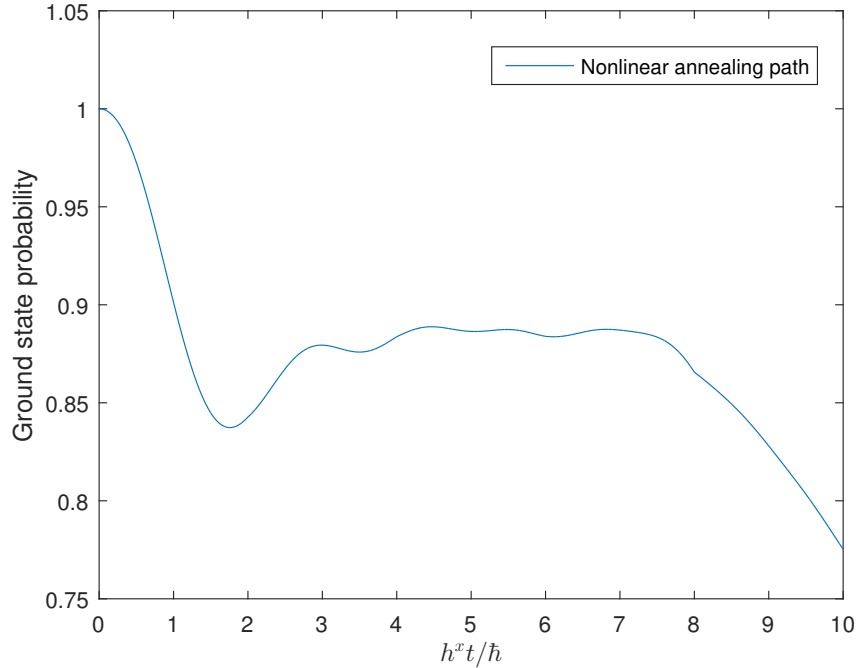


Figure 3.6: How ground state probability evolves with scaled time.

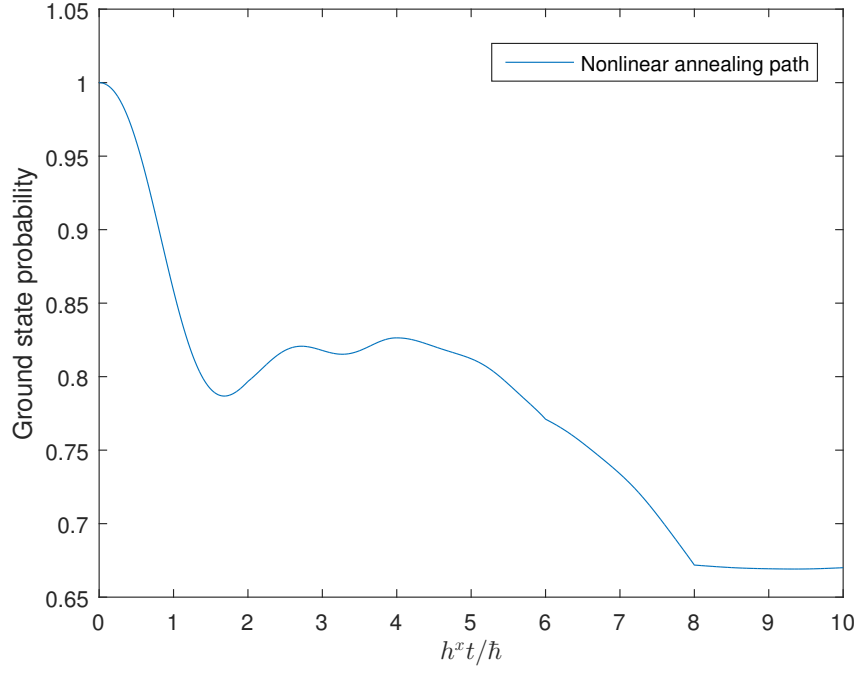


Figure 3.7: How ground state probability evolves with scaled time.

To summarize, this investigation shows that it is possible to get higher ground state probability by choosing the functions $a(t)$ and $b(t)$ in equations (2.5) and (2.6) appropriately with regard to the energy gap. In reality, one does not have the energy spectrum of the problem to be solved; although maybe by doing some mathematical analysis it can be possible to see if there is a typical look of the energy gap for a certain type of problem. Also, note the interesting phenomenon occurring in Figures 3.5, 3.6 and 3.7 that at around 20% of the annealing process, all of the ground state probabilities increased after reaching a local minimum. We expected the ground state probability to remain constant or decrease only slightly when the annealing speed is reduced. This observation implies that, beyond our expectation, somehow the ground state probability “rescued” itself during that time; and perhaps even more interesting is that in Figures 3.6 and 3.7, the “rescue” started *before* speed reduction.

3.5 Ground State Probability versus Annealing Time

According to the adiabatic theorem, the probability of ending in the ground state of $\hat{H}_F$ at $t = t_F$ should converge to 1.0 when $t_F \rightarrow \infty$. In this section, a brief investigation is made regarding how the ground state probability evolves as a function of a finite annealing time t_F , when using a linearly varying annealing path.

Starting off with only two qubits, h_i^z and J_{ij} in the problem Hamiltonian were randomly generated using normal distribution just like in the eigenspectrum investigation. The numerical method using *DHDHD* form is then used to solve the Schrödinger equation. At the end of the annealing process, when $t = t_F$, the probability of the system

being in the ground state is read. In the unlikely case of degenerate ground state energy at the very end of the process, the probabilities of ending up in each ground state are added. By varying the parameter t_F , a plot of ground state probability at $t = t_F$ against the value of t_F is made, see Figure 3.8.

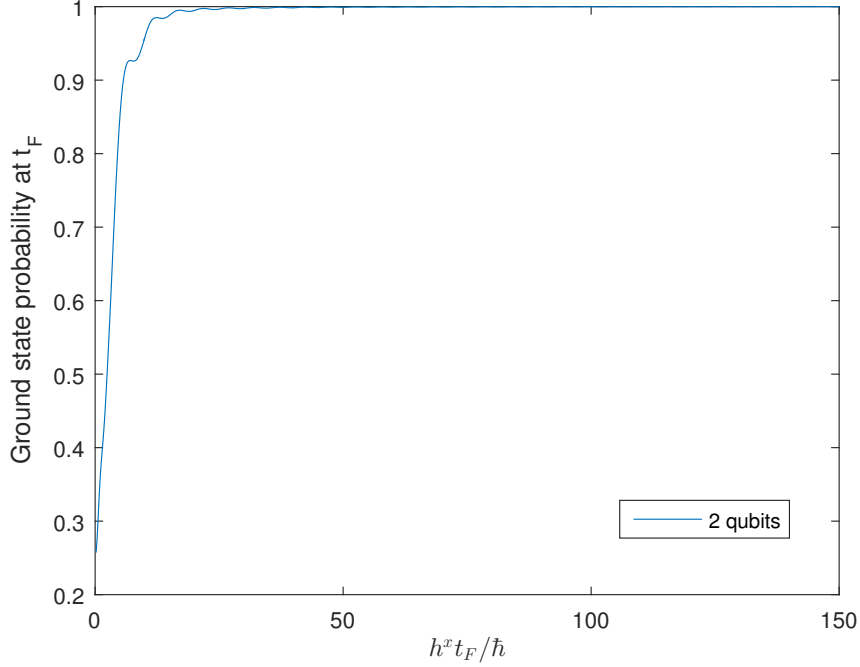


Figure 3.8: The probability of the system ending up in the ground state of the final Hamiltonian as a function of the scaled total annealing time.

A zoom in from the start to the region where the probability goes up to about 0.9 shows a couple of things. First of all, for an infinitely quick process, that is for $t_F = 0$, the probability of the system being in the ground state of $\hat{H}_F$ is 2^{-N} ; the same as measuring any other state. The probability grows almost linearly with t_F up to the value 0.9, see Figure 3.9. A zoom in at the uppermost region in Figure 3.8, where the probability approaches 1.0, shows a peculiar phenomenon. Unlike what we expected, the probability oscillates in a wave form, instead of converging at a steady pace, see Figure 3.10.

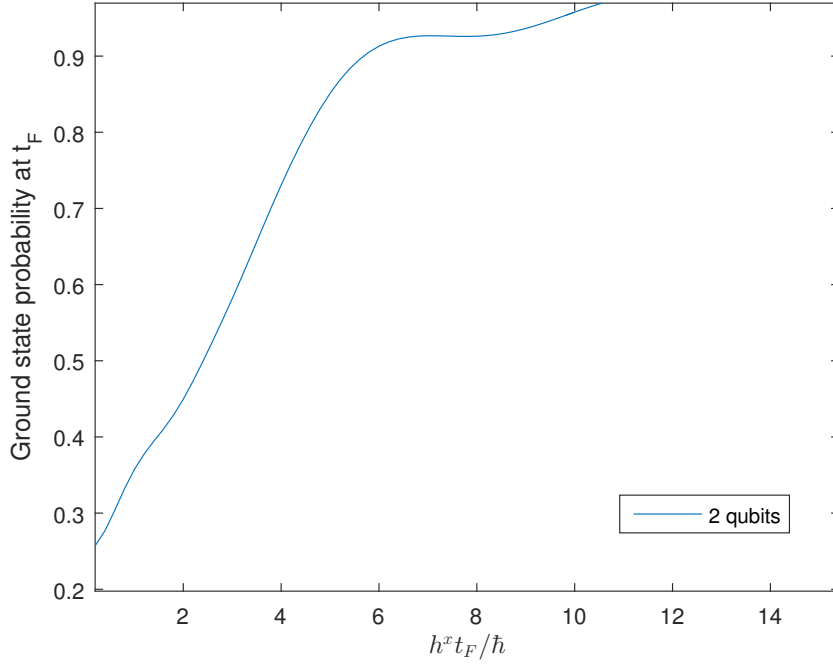


Figure 3.9: The probability of the system ending up in the ground state of the final Hamiltonian as a function of the scaled total annealing time, a zoom in.

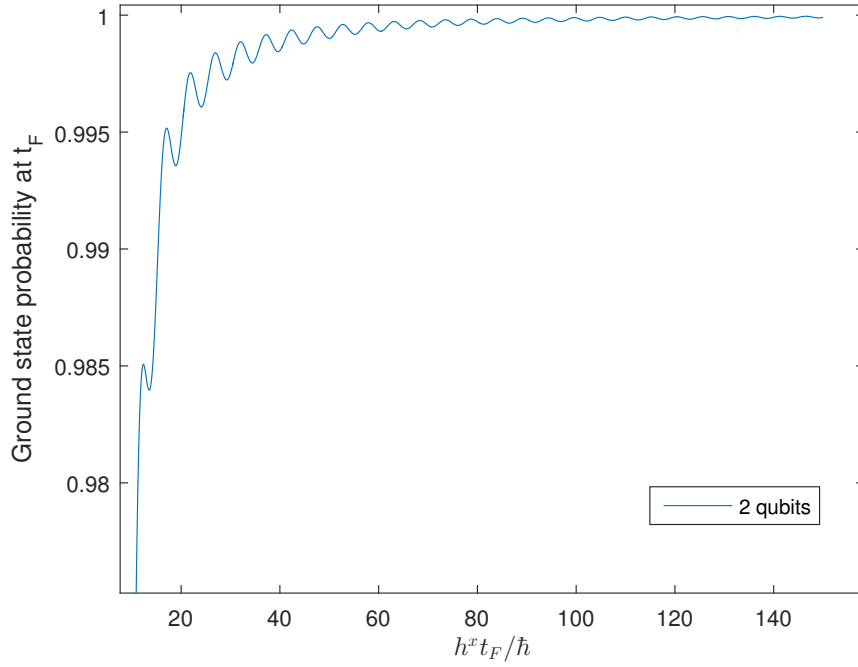


Figure 3.10: The probability of the system ending up in the ground state of the final Hamiltonian as a function of the scaled total annealing time, a zoom in.

A similar plot is made for 8 qubits, see Figure 3.11. With 8 qubits, the probability of the system being in the ground state of $\hat{H}_F$ when $t_F = 0$ is $\frac{1}{2^8} \approx 0.004$, that is, almost

zero, which can be seen on this figure. Comparing the figures, we see that the ground state probability converges to 1.0 in different ways for 2 and 8 qubits. By reducing the value of Δt discussed in section 3.1.3 to achieve a higher numerical accuracy, we could also verify that the wave phenomena that occurred for 2 qubits shown in Figure 3.10 does no longer exist for 8 qubits.

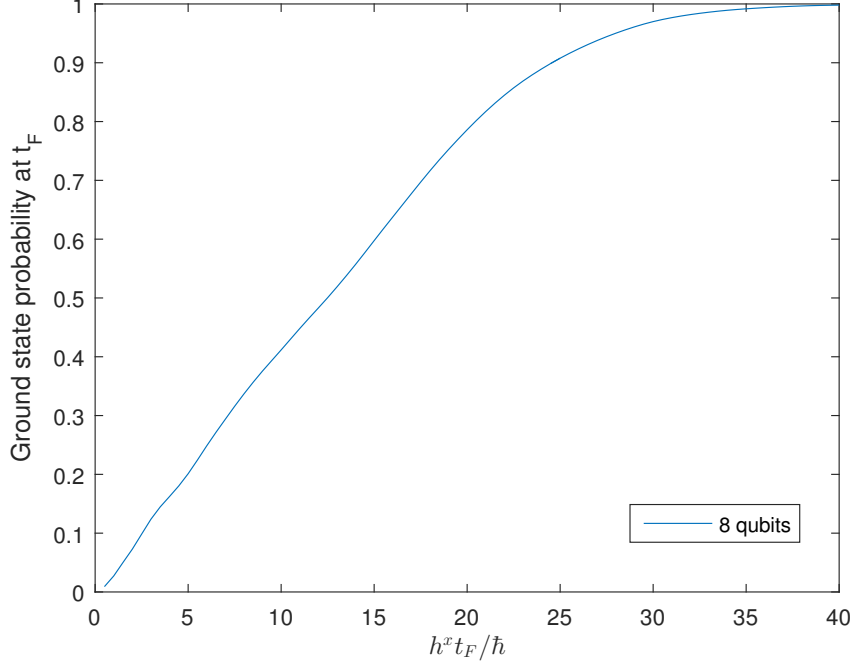


Figure 3.11: The probability of the system ending up in the ground state of the final Hamiltonian as a function of the scaled total annealing time.

To summarize, from the simulation results shown in Figures 3.8 - 3.11, we can see that the ground state probability converges to 1.0 when the annealing time is increased, just as expected. We can also see that the converging rate is different for 2 and 8 qubits; for 2 qubits the probability went up to 0.9 after 6 time units but for 8 qubits it took 25 time units. We investigated this difference in the next section. An interesting and unexpected phenomenon that arose during the investigation is the wave-like appearance of the ground state probability shown in Figure 3.10. We randomized the problem Hamiltonian a couple of times and this phenomena was observed each time.

3.6 Scaling of Qubits

3.6.1 Randomly Generated Problem Hamiltonian

We wanted to investigate how the error in the quantum annealer depend on the number of qubits. Finding patterns for how properties of quantum computers scale with the number of qubits is relevant because in order to solve difficult problems we need more qubits than we can simulate.

We did this by keeping the time of the annealing process constant and randomly creating 200 different problem Hamiltonians with the same method as in the energy spectrum investigation in section 3.3. We ran the simulation using constant rate of change and plotted the average value and standard deviation of measuring the ground state probability for qubit counts between 1 and 11 respectively. The result is shown below in Figure 3.12. We see in Figure 3.12 that problems become increasingly difficult with an increased number of qubits. The semilog-plot in Figure 3.13 shows that the average probability decreases almost exponentially with increasing qubit number. If we look at Eq. (2.9) we see that the sum which is approximated to 0 in the adiabatic approximation will contain an increased number of terms for an increased number of qubits.

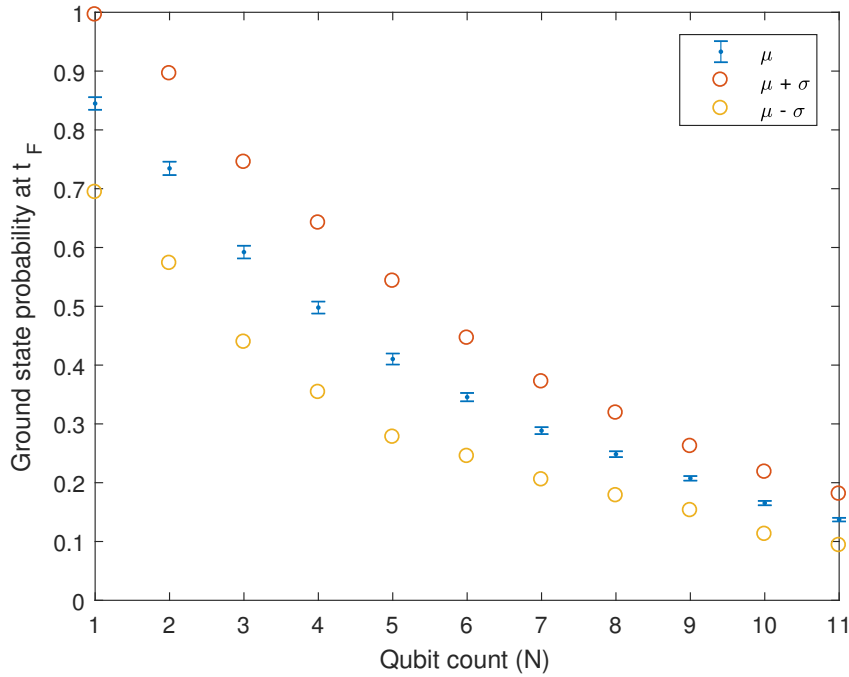


Figure 3.12: μ shows the average value over 200 iterations and σ is the standard deviation.

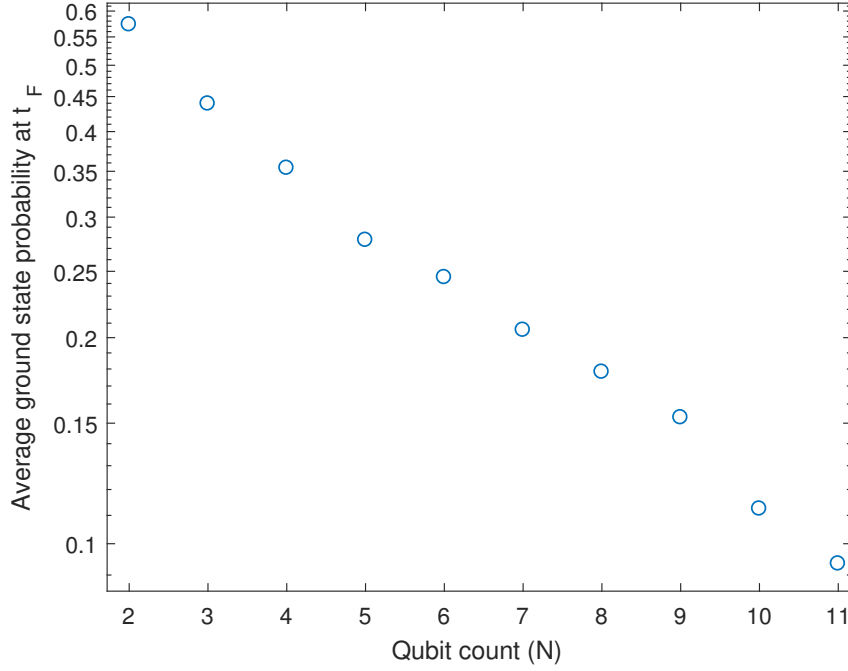


Figure 3.13: The average values μ from Figure 3.12 plotted against qubit count.

The number of states increases as 2^N where N is the number of qubits while the interval of eigenvalues using the Ising model will not increase as quickly. The density of energies will therefore also increase with increasing qubit count.

We wanted to find the amount of time required in order to maintain a constant probability of reaching the correct answer. We aimed at 70% and adjusted the final times $t_F(N)$ which is now a function of the number of qubits N . The times that we used are plotted in Figure 3.14 and the probabilities achieved are shown in Figure 3.15. Note that we skipped the simulation of 11 qubits here since otherwise the simulation will take too long time. We see in Figure 3.15 that the probabilities became pretty close to each other with the exception of the qubit count 2. We therefore choose to only show the times for the qubits 3 - 11 that had approximately similar ground state probability in Figure 3.14.

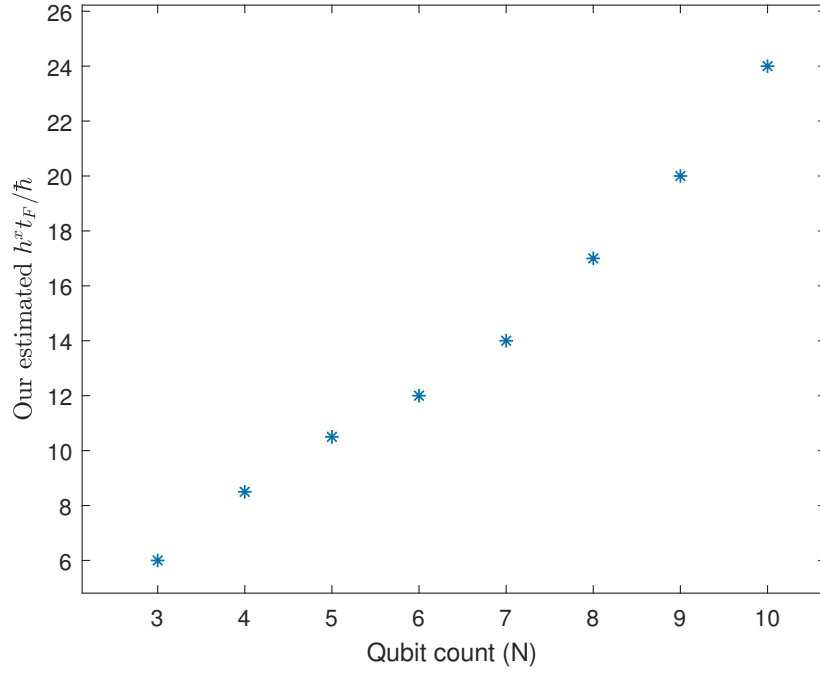


Figure 3.14: Our estimated values of what produces a constant ground state probability. A measurement of how good this guess was is shown in Figure 3.15.

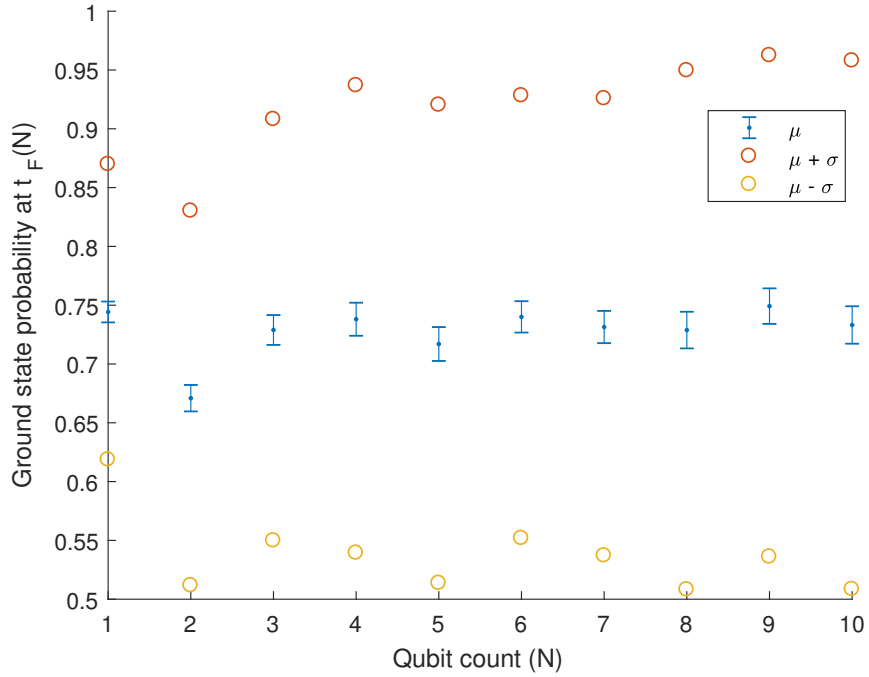


Figure 3.15: Note that the calculation time $t_F(N)$ is now increased with increasing qubit count given by Figure 3.14. μ shows the average value over 200 iterations and σ is the standard deviation.

The purpose of this investigation is to see how the annealing time scales with the

number of qubits. For our 200 randomly generated problem Hamiltonians with normal distribution, a constant speed annealing process and the aim of reaching 70% ground state probability at the end gave the result shown in Figure 3.14. As can be seen, for 3 up to 10 qubits, the annealing time grows almost linearly. Since it is hard to simulate more qubits on a classical computer, we can not tell for sure if this result can be extrapolated to more qubits.

3.6.2 Number Partitioning Problem

In this investigation we wanted to see if the properties we found for randomized constants h_i^z and J_{ij} apply also to the number partitioning problem with problem Hamiltonian taken from Eq. (3.4). For this we randomized the numbers with normal distribution and created the appropriate h_i^z and J_{ij} . We then re-scaled the values in J_{ij} so that the average absolute value is 1. For this problem $h_i^z = 0$ as described in section 3.2. All the values J_{ij} are positive. We choose to do a similar plot as Figure 3.12 by plotting end ground state probability against qubit count. Other than the problem Hamiltonian, we kept all other parameters the same as in the previous investigation. The result is shown in Figure 3.16.

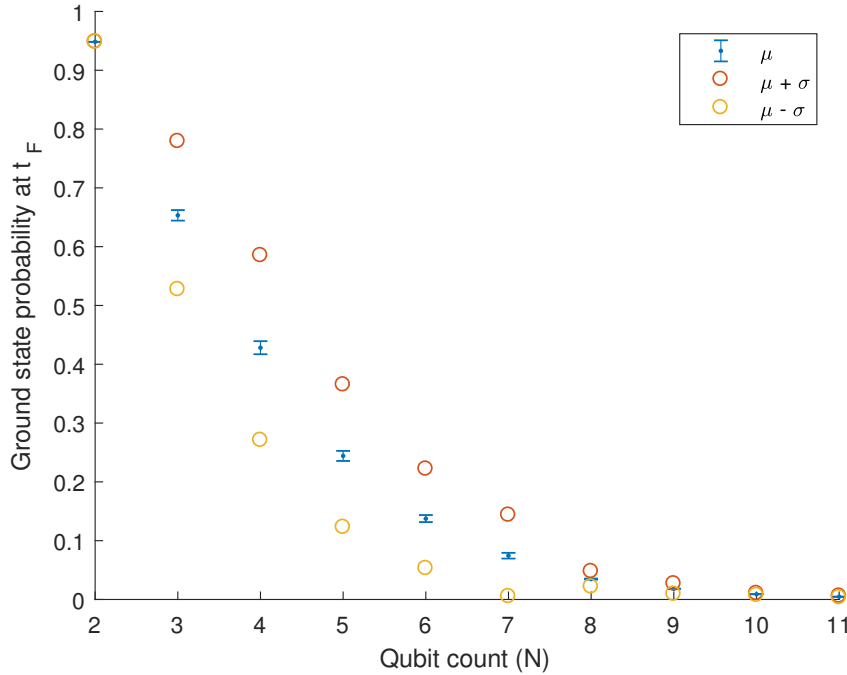


Figure 3.16: μ shows the average value over 200 iterations and σ is the standard deviation.

We see in Figure 3.16 that the probability decreases faster with increasing qubit count compared to Figure 3.12. The reason for this could have been due to $h_i^z = 0$ causing a more dense energy spectrum. Perhaps this is another reason to fixate the first number in the problem into one of the groups; to create non-zero h_i^z parameters in addition to decreasing the number of qubits used. Once again we achieved an exponential decrease seen also by the straight line in the semilog-plot in Figure 3.17.

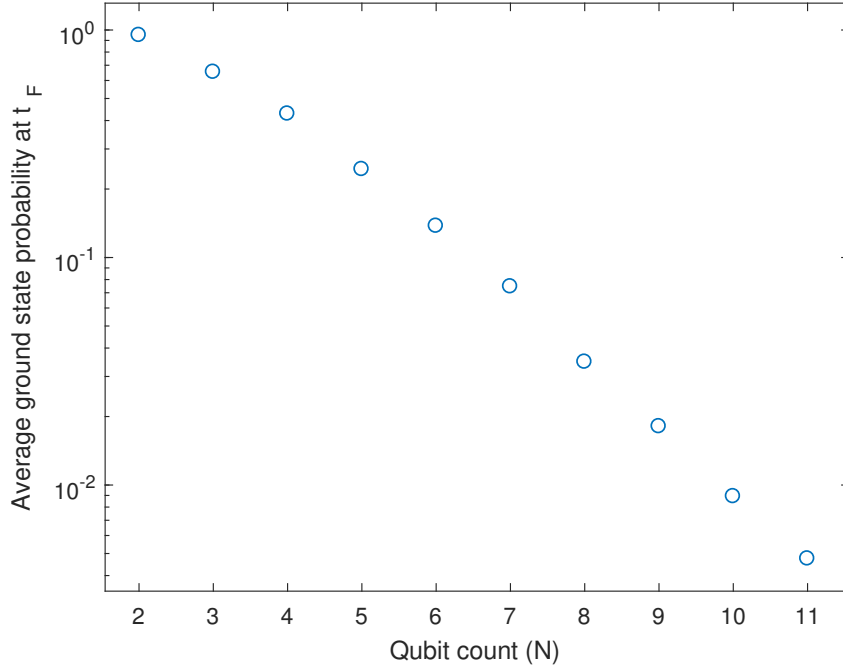


Figure 3.17: The average values μ from Figure 3.16 plotted against qubit count.

In summary we have seen that certain problem Hamiltonians can change the properties of the quantum annealing.

3.7 An Algorithm for Solving the Game Minesweeper

We wrote an algorithm for solving the game minesweeper since it seemed like an appropriate problem for a quantum annealer. We were able to solve it using interaction energies J_{ij} only between qubits that are relatively close to each other in a 2D grid. This is more similar to how real quantum annealers can be built compared to the Ising model with interaction energies between all qubits, and is therefore a big advantage.

Minesweeper consists of a rectangular grid of covered squares. Once uncovered, each square will either hold a single mine or an integer showing the number of mines in the 8 adjacent squares. Using this information, the player's goal is to flag squares that must contain a mine, and uncover squares that cannot contain a mine in order to get more information. The game is lost if a mine is uncovered. The total number of mines on a field is often given, but we will start by disregarding this information.

We choose each qubit to represent a square on the grid, with $s_i = 1$ representing a mine on square with index i and $s_i = -1$ representing an integer on square with index i . Our goal is first to find a configuration of mines that satisfies all known integers on the grid. We let the qubits represent all squares in the grid that are adjacent to an integer. Let n_f represent the number of adjacent flags and n_i are all the known integers. We then

choose the problem Hamiltonian to be

$$H_F = A \sum_{n_i} \left(n_i - n_f - \sum_{\text{adjacent } s_i} \frac{1 + s_i}{2} \right)^2. \quad (3.6)$$

A is here a positive constant. This Hamiltonian is 0 when all known digits are satisfied and greater than 0 otherwise, hence the ground state is a mine configuration that satisfies all known integers if possible. If the square in the Hamiltonian is expanded we reach an expression on Ising form plus a constant which is simply removed. Finding a mine configuration that satisfies all known integers is NP-complete on a classical computer (Kaye 2000).

Our proposed algorithm to solve minesweeper when possible is as follows. Make an assumption that a square of interest is a mine by entering a 1 instead of s_i in the problem Hamiltonian. Then run the quantum annealer and check the result classically. If the given mine configuration does not satisfy all known integers, this means that it is impossible to satisfy all known integers if the ground state was given. This means that the assumption that the square is a mine is incorrect, hence the square cannot contain a mine. However, if a correct mine configuration is returned, we must try assuming a mine or a non-mine in a different square (inputting $s = -1$ for non-mine). If all squares are tested for mines and non-mines and for each case the quantum annealer returns a mine configuration that satisfies all integers we know that the problem cannot be solved.

We now address the fact that the total number of mines can be known. In this case we can let qubits represent all the remaining hidden squares and add the following term to the Hamiltonian where n_m is the known number of mines

$$B \left(n_m - n_f - \sum_{s_i} \frac{1 + s_i}{2} \right)^2. \quad (3.7)$$

B is here a positive constant. This addition of the problem Hamiltonian will unfortunately create interaction energies between all qubits and also require more qubits to be used. Fortunately, it is generally only useful to take this into account when a majority of the uncovered squares are already next to integers since squares that are not adjacent to known integers can be chosen to suit the total number of mines. Additionally we do not run the risk of losing the game by ignoring this information; in the worst case scenario we would instead get stuck in a situation where using this term would help.

We do not necessarily need to look at all squares adjacent to integers in order to gain more information. Therefore, the efficiency might be increased by initially looking only at a few integers at a time. Finally note that the algorithm works in exactly the same way for minesweeper in any dimensions.

Chapter 4

Summary and Conclusions

In our investigations several properties of quantum annealing were illustrated. We explained how an optimization problem can be rewritten to be solved on a quantum annealer and how to simulate a quantum annealer. We found at what part of the quantum annealing the energy gap minimum was typically located for randomized values of energy parameters in the Ising model. With a given energy gap, we verified that it is possible to reach a higher probability of measuring the ground state by choosing an appropriate way of varying the Hamiltonian. We noticed that the ground state probability can “rescue” itself, sometimes even before the annealing speed is reduced. We verified that the probability of measuring the ground state increased for longer annealing times as suggested by the adiabatic approximation, but we also noticed that it was not monotonically increasing, at least not for 2 qubits. We found that in order to reach about 70% probability of measuring the ground state, the annealing time might have to be linearly increasing with increased number of qubits, at least in the region of 3 to 10 qubits. Finally, we wrote our own algorithm for solving the game Minesweeper.

Throughout this report we chose the problem Hamiltonian to be on the Ising form with interaction energies between all qubits even though there are technical difficulties with this for a large number of qubits. A D-wave computer has connections as a chimera graph (Rønnow et al. 2014), so an interesting extension of the project would be investigation of how to write a model with connections adapted to D-wave’s or another quantum annealer. We could have recieved different results using different models and a comparison of these models could be of interest. Another possible extension of the project is to investigate how the problem Hamiltonian typically looks for different problems. If a pattern can be found for a certain type of problem the annealing path can then be optimized for that problem type. Ideally one would find a class of real life problems that are efficiently solved by quantum annealers; this would contribute to the discussion on quantum speedup as well as help with how to best use quantum annealers. It would also be interesting to investigate more in detail our observations that we found counter intuitive, this might lead to a better understanding of the behaviour of quantum systems.

Our work during this project are the first steps in a more ambitious study and the results obtained can be used to find potential properties of larger systems.

Appendices

Appendix A

Details of the Numerical Solution

The linearly time-varying Hamiltonian of the Ising model is

$$\hat{H}(t) = (1 - \frac{t}{t_F})\hat{H}_I + (\frac{t}{t_F})\hat{H}_F = \underbrace{(1 - \frac{t}{t_F})(-h^x \sum_i \sigma_i^x)}_{\equiv \hat{H}^x(t)} + \underbrace{(\frac{t}{t_F})(-\sum_i h_i^z \sigma_i^z - \sum_i \sum_{j>i} J_{ij} \sigma_i^z \sigma_j^z)}_{\equiv \hat{H}^z(t)}. \quad (\text{A.1})$$

In this section the numerical details of solving the Schrödinger equation for this Hamiltonian is carried out.

A couple of concepts are necessary to include for the numerical discretization that follows. The first one is the identity operator expressed as sum of a discrete orthonormal basis $|e_n\rangle$

$$\sum_n |e_n\rangle\langle e_n| = 1. \quad (\text{A.2})$$

The second concept to be included is a modified version of the Trotter formula (Nielsen and Chuang 2000):

$$e^{i(A+B)\Delta t} = e^{iA\Delta t/2} e^{iB\Delta t} e^{iA\Delta t/2} + \mathcal{O}(\Delta t^3), \quad (\text{A.3})$$

where A and B denotes Hermitian operators. Observables are represented by Hermitian operators. $\hat{H}^x(t)$ and $\hat{H}^z(t)$ defined in Eq. (A.1) above are Hermitian operators, because they represent time-varying energies, which are observables. The last thing to be included is the concept of exponential terms with operators as exponent. If k is an arbitrary constant, σ^z is the diagonal 2-dimensional Pauli-z matrix, $e^{k\sigma^z}$ is (Nielsen and Chuang 2000)

$$e^{k\sigma^z} = \begin{bmatrix} e^{k*1} & 0 \\ 0 & e^{k*(-1)} \end{bmatrix} = \begin{bmatrix} e^k & 0 \\ 0 & e^{-k} \end{bmatrix}. \quad (\text{A.4})$$

With these in mind, the numerical solution can now be carried out.

The Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (\text{A.5})$$

The difficulty in making an exact solution is that a time-dependent Hamiltonian does not commute with itself during different times. In other words, Hamiltonian during different

times can not be represented with the same set of eigenbasis. Instead, we have to regard a short time Δt , during which $\hat{H}(t)$ is approximately constant

$$|\psi(t + \Delta t)\rangle \approx e^{-\frac{i}{\hbar}\hat{H}(t)\Delta t}|\psi(t)\rangle. \quad (\text{A.6})$$

For N time intervals Δt we have

$$|\psi(t)\rangle \approx \prod_{n=1}^N e^{-\frac{i}{\hbar}\hat{H}(t_n)\Delta t}|\psi(0)\rangle, \quad (\text{A.7})$$

which becomes exact in the limit $N \rightarrow \infty$ and $\Delta t \rightarrow 0$. Using Eq. (A.1) gives

$$|\psi(t)\rangle \approx \prod_{n=1}^N e^{-\frac{i}{\hbar}(\hat{H}^x(t_n) + \hat{H}^z(t_n))\Delta t}|\psi(0)\rangle. \quad (\text{A.8})$$

We now use the modified Trotter approximation (A.3) to get

$$|\psi(t)\rangle \approx \prod_{n=1}^N e^{-\frac{i}{\hbar}\hat{H}^z(t_n)\frac{\Delta t}{2}} e^{-\frac{i}{\hbar}\hat{H}^x(t_n)\Delta t} e^{-\frac{i}{\hbar}\hat{H}^z(t_n)\frac{\Delta t}{2}}|\psi(0)\rangle. \quad (\text{A.9})$$

We insert the identity operators of $|e_x\rangle$ and $|e_z\rangle$ in Eq. (A.9) according to Eq. (A.2) and look at how the wave function changes from time t to $t + \Delta t$

$$|\psi(t + \Delta t)\rangle \approx e^{-\frac{i}{\hbar}\hat{H}^z\frac{\Delta t}{2}}\left(\sum_{e'_z}|e'_z\rangle\langle e'_z|\right)e^{-\frac{i}{\hbar}\hat{H}^x\Delta t}\left(\sum_{e_x}|e_x\rangle\langle e_x|\right)e^{-\frac{i}{\hbar}\hat{H}^z\frac{\Delta t}{2}}\left(\sum_{e_z}|e_z\rangle\langle e_z|\right)|\psi(t)\rangle. \quad (\text{A.10})$$

We now perform some inner products to get

$$|\psi(t + \Delta t)\rangle \approx \sum_{e'_z e_x e_z} e^{-\frac{i}{\hbar}\hat{H}^z\frac{\Delta t}{2}}|e'_z\rangle\langle e'_z|e^{-\frac{i}{\hbar}\hat{H}^x\Delta t}|e_x\rangle\langle e_x|e^{-\frac{i}{\hbar}\hat{H}^z\frac{\Delta t}{2}}|e_z\rangle\langle e_z|\psi(t)\rangle. \quad (\text{A.11})$$

Note that

$$e^{-\frac{i}{\hbar}\hat{H}^x\Delta t}|e_x\rangle = e^{-\frac{i}{\hbar}(1-\frac{t}{t_F})(-h^x\sum_i\sigma_i^x)\Delta t}|e_x\rangle = e^{-\frac{i}{\hbar}(1-\frac{t}{t_F})(-h^x\sum_i s_i^x)\Delta t}|e_x\rangle \quad (\text{A.12})$$

$$\begin{aligned} e^{-\frac{i}{\hbar}\hat{H}^z\frac{\Delta t}{2}}|e_z\rangle &= e^{-\frac{i}{\hbar}(\frac{t}{t_F})(-\sum_i h_i^z\sigma_i^z - \sum_i \sum_{j>i} J_{ij}\sigma_i^z\sigma_j^z)\frac{\Delta t}{2}}|e_z\rangle \\ &= e^{-\frac{i}{\hbar}(\frac{t}{t_F})(-\sum_i h_i^z s_i^z - \sum_i \sum_{j>i} J_{ij}s_i^z s_j^z)\frac{\Delta t}{2}}|e_z\rangle. \end{aligned} \quad (\text{A.13})$$

There are different ways of expressing an operator, either with their eigenbasis or with any other orthonormal basis. In matrix representation, any operator expressed in their eigenbasis, comes out as a diagonal matrix. Therefore, the trick here is to express both the Pauli-x matrix and the Pauli-z matrix in their eigenbasis. This will allow us to achieve faster multiplication when implemented in MATLAB. Putting these intermediate results (A.12) and (A.13) into Eq. (A.11) gives

$$\begin{aligned} |\psi(t + \Delta t)\rangle &\approx \sum_{e'_z} |e'_z\rangle e^{-\frac{i}{\hbar}(\frac{t_n}{t_F})(-\sum_i h_i^z s_i^{z'} - \sum_i \sum_{j>i} J_{ij}s_i^{z'} s_j^{z'})\frac{\Delta t}{2}} \sum_{e_x} \langle e'_z|e_x\rangle e^{-\frac{i}{\hbar}(1-\frac{t_n}{t_F})(-h^x\sum_i s_i^x)\Delta t} \\ &\quad \sum_{e_z} \langle e_x|e_z\rangle e^{-\frac{i}{\hbar}(\frac{t_n}{t_F})(-\sum_i h_i^z s_i^z - \sum_i \sum_{j>i} J_{ij}s_i^z s_j^z)\frac{\Delta t}{2}} \psi_{e_z}(t). \end{aligned} \quad (\text{A.14})$$

The result (A.14) includes two sums of inner products

$$\begin{aligned}
\langle e_x | e_z \rangle &= \langle e_x^1 e_x^2 \dots e_x^m | e_z^1 e_z^2 \dots e_z^m \rangle \\
&= (\langle e_x^1 | \otimes \langle e_x^2 | \otimes \dots \otimes \langle e_x^m |) (| e_z^1 \rangle \otimes | e_z^2 \rangle \otimes \dots \otimes | e_z^m \rangle) \\
&= \langle e_x^1 | e_z^1 \rangle \langle e_x^2 | e_z^2 \rangle \dots \langle e_x^m | e_z^m \rangle \\
&= H_{e_x e_z},
\end{aligned} \tag{A.15}$$

where H is known as a Hadamard matrix and it serves as a basis change function. When H acts on $\psi_{e_z}(0)$, it changes the representation of the wave function at initial time, from being expressed in the z -basis to being expressed in the x -basis. In a Hilbert space of dimension 2^m , a Hadamard matrix has the dimension $2^m \times 2^m$.

To summarize, each time step consists of matrix multiplications with diagonal (D) and Hadamard (H) matrices in the form $DHDDH$.

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